Package 'Satisficing'

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Type Package

Title Tools for Detecting Indicators of Survey Satisficing

Version 0.3.0

Description Provides tools for detecting potential indicators of satisficing in survey responses, including straight-line response patterns and row-wise evaluation of logical conditions, as well as careless and insufficient effort response indices ported from the 'careless' package (including longstring, intra-individual response variability, even-odd consistency, psychometric synonyms/antonyms, and Mahalanobis distance).

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URL https://github.com/soheidon/Satisficing

BugReports https://github.com/soheidon/Satisficing/issues

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Contents

careless_dataset 2
careless_dataset2 2
evenodd 3
irv 4
longstring 5
mahad 6
matching 7
maxIdenticalRating 7
meanRootPairs 8
mir 10

mrp . . . 11
nondiff . . . 12
print.straightliner . . . 13
psychant . . . 13
psychsyn . . . 14
psychsyn_critval . . . 15
question_sd . . . 16
rawnn . . . 17
scale_point_variation . . . 17
spv . . . 18
stline . . . 19
straightlining . . . 20
straightlining_qset . . . 21

Index **23**

`careless_dataset` *Simulated dataset with insufficient effort responses.*

Description

A simulated dataset mimicking insufficient effort responding. Contains three types of responses: (a) Normal responses with answers centering around a trait/attitude value (80 percent probability per simulated observation), (b) Straightlining responses (10 percent probability per simulated observation), (c) Random responses (10 percent probability per simulated observation). Simulated are 10 subscales of 5 items each (= 50 variables). Ported from the **careless** package (version 1.2.2).

Usage

```
careless_dataset
```

Format

A data frame with 200 observations (rows) and 50 variables (columns).

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

`careless_dataset2` *Simulated dataset with careless responses.*

Description

A simulated dataset mimicking insufficient effort responding. Contains three types of responses: (a) Normal responses with answers mimicking a diligent respondent (b) Some number of longstring careless responders, (c) some number of generally careless responders. Simulated are 10 subscales of 10 items each (= 100 variables). Ported from the **careless** package (version 1.2.2).

Usage

```
careless_dataset2
```

Format

A data frame with 1000 observations (rows) and 100 variables (columns).

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

---

evenodd *Calculates the even-odd consistency score*

Description

Takes a matrix of item responses and a vector of integers representing the length each factor. The even-odd consistency score is then computed as the within-person correlation between the even and odd subscales over all the factors. Ported from the **careless** package (version 1.2.2).

Usage

```
evenodd(x, factors, diag = FALSE)
```

Arguments

`x`	a matrix or data frame of data (e.g. survey responses)
`factors`	a vector of integers specifying the length of each factor in the dataset
`diag`	optionally returns a column with the number of available (i.e., non-missing) even/odd pairs per observation. Useful for datasets with many missing values.

Value

If `diag = FALSE`, a numeric vector of even-odd consistency scores. If `diag = TRUE`, a data frame containing `eo_cor` and `eo_missing`.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Johnson, J. A. (2005). Ascertaining the validity of individual protocols from web-based personality inventories. *Journal of Research in Personality, 39*, 103-129. doi:10.1016/j.jrp.2004.09.009

Examples

```
careless_eo <- evenodd(careless_dataset, rep(5,10))
careless_eodiag <- evenodd(careless_dataset, rep(5,10), diag = TRUE)
```

`irv` *Calculates the intra-individual response variability (IRV)*

Description

The IRV is the "standard deviation of responses across a set of consecutive item responses for an individual" (Dunn, Heggestad, Shanock, & Theilgard, 2018, p. 108). By default, the IRV is calculated across all columns of the input data. Additionally it can be applied to different subsets of the data. This can detect degraded response quality which occurs only in a certain section of the questionnaire (usually the end). With `split = FALSE`, IRV is the standard deviation across the selected response values and is invariant to their order. With `split = TRUE`, consecutive column subsets are formed in their current order, so reordering columns can change subset membership and subset IRV values. Unlike `longstring`, `irv()` does not measure runs of consecutive identical responses. Whereas Dunn et al. (2018) propose to mark persons with *low* IRV scores as outliers - reflecting straightlining responses, Marjanovic et al. (2015) propose to mark persons with *high* IRV scores - reflecting highly random responses (see References). Ported from the **careless** package (version 1.2.2).

Usage

```
irv(x, na.rm = TRUE, split = FALSE, num.split = 3)
```

Arguments

`x`	a matrix or data frame of data (e.g. survey responses)
`na.rm`	logical indicating whether to calculate the IRV for a person with missing values.
`split`	logical indicating whether to additionally calculate the IRV on subsets of columns (of equal length).
`num.split`	the number of subsets the data is to be split in.

Value

If `split = FALSE`, a numeric vector of IRV values. If `split = TRUE`, a data frame containing `irvTotal` and subset columns (`irv1`, `irv2`, etc.).

Author(s)

Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Dunn, A. M., Heggestad, E. D., Shanock, L. R., & Theilgard, N. (2018). Intra-individual Response Variability as an Indicator of Insufficient Effort Responding: Comparison to Other Indicators and Relationships with Individual Differences. *Journal of Business and Psychology, 33(1)*, 105-121. doi:10.1007/s1086901694790

Marjanovic, Z., Holden, R., Struthers, W., Cribbie, R., & Greenglass, E. (2015). The inter-item standard deviation (ISD): An index that discriminates between conscientious and random responders. *Personality and Individual Differences*, 84, 79-83. doi:10.1016/j.paid.2014.08.021

Examples

```
# calculate the irv over all items
irv_total <- irv(careless_dataset)

# calculate the irv over all items + calculate the irv for each quarter of the questionnaire
irv_split <- irv(careless_dataset, split = TRUE, num.split = 4)
boxplot(irv_split$irv4) #produce a boxplot of the IRV for the fourth quarter
```

---

longstring *Identifies the longest string of identical consecutive responses for each observation*

Description

Takes a matrix of item responses and, beginning with the second column (i.e., second item) compares each column with the previous one to check for matching responses. For each observation, the length of the maximum uninterrupted string of identical responses is returned. Additionally, can return the average length of uninterrupted string of identical responses. Ported from the **careless** package (version 1.2.2).

Usage

```
longstring(x, avg = FALSE)
```

Arguments

x	a matrix or data frame of data (e.g. item responses)
avg	logical indicating whether to additionally return the average length of identical consecutive responses

Value

If `avg = FALSE`, a vector of lengths of the longest uninterrupted string of identical responses. If `avg = TRUE`, a data frame containing `longstr` and `avgstr`.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Johnson, J. A. (2005). Ascertaining the validity of individual protocols from web-based personality inventories. *Journal of Research in Personality, 39*, 103-129. doi:10.1016/j.jrp.2004.09.009

Examples

```
careless_long <- longstring(careless_dataset, avg = FALSE)
careless_avg <- longstring(careless_dataset, avg = TRUE)
boxplot(careless_avg$longstr) #produce a boxplot of the longstring index
boxplot(careless_avg$avgstr)
```

mahad *Find and graph Mahalanobis Distance (D) and flag potential outliers.*

Description

Takes a matrix of item responses and computes Mahalanobis D. Can additionally return a vector of binary outlier flags. Mahalanobis distance is calculated using the function `psych::outlier` of the **psych** package, an implementation which supports missing values. Ported from the **careless** package (version 1.2.2).

Usage

```
mahad(x, plot = TRUE, flag = FALSE, confidence = 0.99, na.rm = TRUE)
```

Arguments

`x`	a matrix or data frame of data
`plot`	Plot the resulting QQ graph
`flag`	Flag potential outliers using the confidence level specified in parameter `confidence`
`confidence`	The desired confidence level of the result
`na.rm`	Should missing data be deleted

Value

If `flag = FALSE`, a numeric vector of squared Mahalanobis distances (`d_sq`). If `flag = TRUE`, a data frame containing `d_sq` and boolean `flagged`.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods, 17(3)*, 437-455. doi:10.1037/a0028085

See Also

`psych::outlier` on which this function is based.

Examples

```
mahad_raw <- mahad(careless_dataset) #only the distances themselves
mahad_flags <- mahad(careless_dataset, flag = TRUE) #additionally flag outliers
mahad_flags <- mahad(careless_dataset, flag = TRUE, confidence = 0.999) #Apply a strict criterion
```

matching *Identify Rows Matching a Logical Condition*

Description

Evaluates a logical condition within the context of a data frame and adds an integer column indicating which rows satisfy the condition.

Usage

```
matching(data, condition, result_col = "match")
```

Arguments

data	A data frame.
condition	An expression to evaluate within data. Must evaluate to a logical vector whose length is exactly nrow(data).
result_col	A non-empty character string specifying the name of the result column. Defaults to "match". Must not conflict with existing columns in data.

Value

The input data frame with an additional integer column indicating whether each row satisfies the condition:

- 1L: The condition evaluated to TRUE.
- 0L: The condition evaluated to FALSE.
- NA_integer_: The condition evaluated to NA.

Examples

```
df <- data.frame(ID = c(1, 2, 3, 4), Value = c(10, 20, 30, 40))
matching(df, Value > 20)
```

maxIdenticalRating *Maximum Identical Rating*

Description

Calculates the proportion of answers taken up by the most common response by the respondent to a set of questions (Kim et al., 2019).

Usage

```
maxIdenticalRating(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

x	A data frame or tibble containing survey responses.
varnames	A vector of variable names or column indices. If NULL (default), all columns of x are used.
verbose	Logical; if TRUE (default), outputs informational messages such as how missing data are handled.
clean_data	Logical; should non-numeric columns be converted to numeric? Defaults to TRUE.
round_digits	A numeric value specifying digits for rounding (kept for API compatibility).

Value

A numeric vector of the same length as nrow(x) indicating the proportion of answers that take the most common value.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5),
  item2 = c(5, 5),
  item3 = c(5, 5),
  item4 = c(5, 1)
)
maxIdenticalRating(survey, varnames = c("item1", "item4"), verbose = FALSE)
```

---

meanRootPairs *Mean Root Pairs*

Description

Computes the sample-normalized Mean Root of Pairs (MRP) straightlining measure for each respondent across a battery of questions (Kim et al., 2019).

Usage

```
meanRootPairs(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

x	A data frame or tibble containing survey responses.
varnames	A vector of variable names or column indices. If NULL (default), all columns of x are used.
verbose	Logical; if TRUE (default), outputs informational messages.
clean_data	Logical; should non-numeric columns be converted to numeric? Defaults to TRUE.
round_digits	A numeric value specifying digits for rounding (kept for API compatibility).

Details

meanRootPairs() strictly implements the mathematical definition from Kim et al. (2019):

$$T_i = \frac{2}{n_i(n_i-1)} \sum_{j<k, j,k \in A_i} \sqrt{|x_{ij} - x_{ik}|}$$

for respondents with at least 2 valid responses, followed by sample-level normalization:

$$\text{MRP}_i = \frac{\max(T) - T_i}{\max(T) - \min(T)}$$

If fewer than 2 respondents have valid T_i values, or if $\max(T) == \min(T)$, NA_real_ is returned for all rows.

Value

A numeric vector of length nrow(x) containing the normalized MRP scores, where higher values represent relatively greater straightlining (lower differentiation).

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(1, 1, 3, 1),
  item2 = c(1, 2, 4, 1),
  item3 = c(1, 3, 5, 2)
)
meanRootPairs(survey, varnames = c("item1", "item2", "item3"), verbose = FALSE)
```

mir *Maximum Identical Rating (MIR)*

Description

Computes the Maximum Identical Rating (MIR) straightlining measure across specified columns for each row in a data frame, as described by Kim et al. (2019).

Usage

```
mir(data, columns, result_column_name = "mir")
```

Arguments

`data`	A data frame.
`columns`	A character vector containing at least 2 unique column names existing in `data`. All specified columns must be numeric (integer/double) or all must be factors.
`result_column_name`	A non-empty character string of length 1 specifying the name of the result column to append. Defaults to `"mir"`. Must not already exist in `data`.

Details

`mir()` is a backward-compatibility wrapper around `maxIdenticalRating`. It appends a result column to the input data frame, computing the proportion of answers taking the most common response value across the specified columns.

Value

The original data frame `data` with an additional numeric column named by `result_column_name` containing the MIR scores.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
df <- data.frame(
  q1 = c(1, 1, 3),
  q2 = c(1, 2, 3),
  q3 = c(1, 3, 3)
)
mir(df, c("q1", "q2", "q3"))
```

mrp *Mean Root of Pairs (MRP)*

Description

Computes the Mean Root of Pairs (MRP) straightlining measure across specified columns for each row in a data frame, as described by Kim et al. (2019).

Usage

```
mrp(data, columns, result_column_name = "mrp")
```

Arguments

data	A data frame.
columns	A character vector containing at least 2 unique column names existing in data. Columns must be numeric (integer or double). Factor, character, or logical columns are not permitted.
result_column_name	A non-empty character string of length 1 specifying the name of the result column to append. Defaults to "mrp". Must not already exist in data.

Details

mrp() is a backward-compatibility wrapper around meanRootPairs. It appends a result column to the input data frame, computing the sample-normalized Mean Root of Pairs across the specified numeric columns.

Value

The original data frame data with an additional numeric column named by result_column_name containing the normalized MRP scores.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
df <- data.frame(
  q1 = c(1, 1, 3, 1),
  q2 = c(1, 2, 4, 1),
  q3 = c(1, 3, 5, 2)
)
mrp(df, c("q1", "q2", "q3"))
```

nondiff *Simple Nondifferentiation Indicator*

Description

Evaluates whether a respondent exhibits complete nondifferentiation across a battery of questions (Kim et al., 2019).

Usage

```
nondiff(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

x	A data frame or tibble containing survey responses.
varnames	A vector of variable names or column indices. If NULL (default), all columns of x are used.
verbose	Logical; if TRUE (default), prints the proportion of nondifferentiation.
clean_data	Logical; should non-numeric columns be converted to numeric? Defaults to TRUE.
round_digits	A numeric value specifying digits for rounding printed output.

Details

In accordance with straightliner conventions, `nondiff()` evaluates whether `length(unique(row)) == 1L`. If a row consists entirely of missing values (NA), `unique()` has length 1, evaluating to TRUE. If a row has both observed values and NA, `unique()` has length greater than 1, evaluating to FALSE.

Note that this differs from `stline`, which returns an integer flag (`1L`, `0L`, or `NA_integer_`) and evaluates whether all non-missing values are identical when at least 2 non-missing responses are present.

Value

A logical vector indicating whether each respondent gave identical answers to all specified questions (TRUE) or not (FALSE).

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5),
  item2 = c(5, 5),
  item3 = c(5, 5),
  item4 = c(5, 1)
)
nondiff(survey, varnames = c("item1", "item4"), verbose = FALSE)
```

---

print.straightliner *Print Method for straightliner Objects*

Description

Print Method for straightliner Objects

Usage

```
## S3 method for class 'straightliner'
print(x, digits = 2, ...)
```

Arguments

x	An object of class straightliner.
digits	Number of decimal places for rounding output.
...	Additional arguments passed to print.

---

psychant *Computes the psychometric antonym score*

Description

A convenient wrapper that calls psychsyn with argument anto = TRUE to compute the psychometric antonym score. Ported from the **careless** package (version 1.2.2).

Usage

```
psychant(x, critval = -0.6, diag = FALSE)
```

Arguments

x	is a matrix or data frame of item responses
critval	is the minimum magnitude of the correlation between two items in order for them to be considered psychometric synonyms. Defaults to -.60
diag	additionally return the number of item pairs available for each subject. Useful if dataset contains many missing values.

Value

If diag = FALSE, a numeric vector of psychometric antonym scores. If diag = TRUE, a data frame containing numPairs and cor.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

See Also

psychsyn for the main function, psychsyn_critval for a helper that allows to set an adequate critical value for the size of the correlation.

Examples

```
antonyms <- psychant(careless_dataset2, .50)
antonyms <- psychant(careless_dataset2, .50, diag = TRUE)
```

---

psychsyn *Computes the psychometric synonym/antonym score*

Description

Takes a matrix of item responses and identifies item pairs that are highly correlated within the overall dataset. What defines "highly correlated" is set by the critical value (e.g., r > .60). Each respondents' psychometric synonym score is then computed as the within-person correlation between the identified item-pairs. Alternatively computes the psychometric antonym score which is a variant that uses item pairs that are highly *negatively* correlated. Ported from the **careless** package (version 1.2.2).

Usage

```
psychsyn(x, critval = 0.6, anto = FALSE, diag = FALSE, resample_na = TRUE)
```

Arguments

x	is a matrix or data frame of item responses
critval	is the minimum magnitude of the correlation between two items in order for them to be considered psychometric synonyms. Defaults to .60
anto	determines whether psychometric antonyms are returned instead of psychometric synonyms. Defaults to FALSE
diag	additionally return the number of item pairs available for each observation. Useful if dataset contains many missing values.
resample_na	if psychsyn returns NA for a respondent resample to attempt getting a non-NA result.

Value

If diag = FALSE, a numeric vector of psychometric synonym scores (correlations). If diag = TRUE, a data frame containing numPairs and cor.

Author(s)

Richard Yentes <ryentes@gmail.com>, Francisco Wilhelm <franciscowilhelm@gmail.com>

References

Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods, 17(3)*, 437-455. doi:10.1037/a0028085

See Also

psychant for a more concise way to calculate the psychometric antonym score, psychsyn_critval for a helper that allows to set an adequate critical value for the size of the correlation.

Examples

```
synonyms <- psychsyn(careless_dataset, .60)
antonyms <- psychsyn(careless_dataset2, .50, anto = TRUE)
antonyms <- psychant(careless_dataset2, .50)

#with diagnostics
synonyms <- psychsyn(careless_dataset, .60, diag = TRUE)
antonyms <- psychant(careless_dataset2, .50, diag = TRUE)
```

---

psychsyn_critval *Compute the correlations between all possible item pairs and order them by the magnitude of the correlation*

Description

A function intended to help finding adequate critical values for psychsyn and psychant. Takes a matrix of item responses and returns a data frame giving the correlations of all item pairs ordered by the magnitude of the correlation. Ported from the **careless** package (version 1.2.2).

Usage

```
psychsyn_critval(x, anto = FALSE)
```

Arguments

x	a matrix or data frame of item responses.
anto	ordered by the largest positive correlation, or, if anto = TRUE, the largest negative correlation.

Value

A data frame with columns var1, var2, and cor, sorted by correlation magnitude.

Author(s)

Francisco Wilhelm <franciscowilhelm@gmail.com>

See Also

after determining an adequate critical value, continue with psychsyn and/or psychant

Examples

```
psychsyn_cor <- psychsyn_critval(careless_dataset)
psychsyn_cor <- psychsyn_critval(careless_dataset, anto = TRUE)
```

---

question_sd *Standard Deviation across Question Battery*

Description

Computes the intra-individual standard deviation across a set of questions for each respondent (Kim et al., 2019).

Usage

```
question_sd(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

x	A data frame or tibble containing survey responses.
varnames	A vector of variable names or column indices. If NULL (default), all columns of x are used.
verbose	Logical; if TRUE (default), outputs informational messages regarding missing data handling.
clean_data	Logical; should non-numeric columns be converted to numeric? Defaults to TRUE.
round_digits	A numeric value specifying digits for rounding (kept for API compatibility).

Value

A numeric vector of standard deviations for each respondent across the specified questions.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5),
  item2 = c(5, 5),
  item3 = c(5, 5),
  item4 = c(5, 1)
)
question_sd(survey, varnames = c("item1", "item3", "item4"), verbose = FALSE)
```

---

rawnn *Print Column Names with Indices*

Description

Prints the column names of a data frame along with their respective 1-based indices, formatted as `'index: column_name'`.

Usage

```
rawnn(dataframe)
```

Arguments

`dataframe`	A data frame whose column names are to be printed.

Value

`invisible(NULL)`. Called for its side effect of printing.

Examples

```
rawnn(mtcars)
```

---

scale_point_variation *Scale Point Variation*

Description

Computes the Scale Point Variation measure capturing response diversity across questions for each respondent (Kim et al., 2019).

Usage

```
scale_point_variation(
  x,
  varnames = NULL,
  verbose = TRUE,
  clean_data = TRUE,
  round_digits = 2
)
```

Arguments

x	A data frame or tibble containing survey responses.
varnames	A vector of variable names or column indices. If NULL (default), all columns of x are used.
verbose	Logical; if TRUE (default), outputs informational messages, including the battery-level statistic and handling of missing data.
clean_data	Logical; should non-numeric columns be converted to numeric? Defaults to TRUE.
round_digits	A numeric value specifying digits for rounding informational messages.

Value

A numeric vector of length nrow(x) with Scale Point Variation scores.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5),
  item2 = c(5, 5),
  item3 = c(5, 5),
  item4 = c(5, 1)
)
scale_point_variation(survey, varnames = c("item1", "item4"), verbose = FALSE)
```

---

spv *Scale Point Variation (SPV)*

Description

Computes the Scale Point Variation (SPV) straightlining measure across specified columns for each row in a data frame, as described by Kim et al. (2019).

Usage

```
spv(data, columns, result_column_name = "spv")
```

Arguments

data	A data frame.
columns	A character vector containing at least 2 unique column names existing in data. All specified columns must be numeric (integer/double) or all must be factors.
result_column_name	A non-empty character string of length 1 specifying the name of the result column to append. Defaults to "spv". Must not already exist in data.

Details

spv() is a backward-compatibility wrapper around scale_point_variation. It appends a result column to the input data frame, computing response diversity across categories for the specified columns.

Value

The original data frame data with an additional numeric column named by result_column_name containing the SPV scores.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
df <- data.frame(
  q1 = c(1, 1, 3),
  q2 = c(1, 2, 3),
  q3 = c(1, 3, 3)
)
spv(df, c("q1", "q2", "q3"))
```

---

stline *Check for Straight-Line Responses Across Specified Columns*

Description

Evaluates each row across specified columns in a data frame to identify straight-line responses (identical answers across items).

Usage

```
stline(
  data,
  columns,
  result_column_name = "straight_line",
  exclude_values = NULL
)
```

Arguments

data	A data frame. Must not contain duplicate column names.
columns	A non-empty character vector of unique column names existing in data.
result_column_name	A non-empty character string specifying the name of the result column. Defaults to "straight_line". Must not conflict with existing columns in data.
exclude_values	An optional atomic vector of values to exclude from straight-line detection. If identical values match any value in exclude_values, the result is 0. Defaults to NULL.

Value

The input data frame with an additional integer column indicating whether the row is a straight-line response:

- `1L`: At least two non-missing values are identical and not excluded.
- `0L`: Non-missing values vary across columns, or identical values match `exclude_values`.
- `NA_integer_`: The row has 0 or 1 non-missing value.

Examples

```
df <- data.frame(
  q1 = c(1, 1, 1, 2, NA),
  q2 = c(1, 1, 2, 2, NA),
  q3 = c(1, 2, 3, 2, 1)
)
stline(df, c("q1", "q2", "q3"))
```

`straightlining` *Individual-Level Straightlining Indicators*

Description

Calculates individual-level indicators of straightlining across a specified set of columns in a survey data frame (Kim et al., 2019).

Usage

```
straightlining(
  x,
  varnames = NULL,
  measures = c("mrp", "mir", "qsd", "spv", "nondiff"),
  keep_original = FALSE
)
```

Arguments

`x`	A data frame or tibble containing survey responses.
`varnames`	A vector of variable names or column indices to be analyzed. If `NULL` (default), all columns of `x` are used.
`measures`	A character vector of the measures to compute. Valid options are: • `"mrp"`: Mean Root of Pairs (`meanRootPairs`). • `"mir"`: Maximum Identical Rating (`maxIdenticalRating`). • `"qsd"`: Battery Standard Deviation (`question_sd`). • `"spv"`: Scale Point Variation (`scale_point_variation`). • `"nondiff"`: Simple Nondifferentiation (`nondiff`). Defaults to `c("mrp", "mir", "qsd", "spv", "nondiff")`.
`keep_original`	Logical; if `TRUE`, returns the original data frame with the computed straightlining indicator columns appended on the right. If `FALSE` (default), returns only the requested straightlining measures as a data frame.

Value

A data frame containing one column per requested measure. If `keep_original = TRUE`, all original columns in `x` are also included.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5, 3),
  item2 = c(5, 5, 2),
  item3 = c(5, 5, 1),
  item4 = c(5, 1, 4)
)
straightlining(survey, varnames = paste0("item", 1:4),
               measures = c("mir", "qsd", "spv", "nondiff"))
```

`straightlining_qset` *Aggregate Straightlining Measures for Question Batteries*

Description

Computes aggregate (battery-level) straightlining measures across multiple sets of questions (Kim et al., 2019).

Usage

```
straightlining_qset(
  x,
  variable_list = NULL,
  measures = c("mrp", "mir", "qsd", "spv", "nondiff"),
  round_digits = 2,
  verbose = FALSE
)
```

Arguments

`x`	A data frame or tibble containing survey responses.
`variable_list`	A named list where each element is a vector containing the column names or column indices for one battery of questions.
`measures`	A character vector of the measures to compute. Valid options are: `"mrp"`, `"mir"`, `"qsd"`, `"spv"`, and `"nondiff"`. Defaults to `c("mrp", "mir", "qsd", "spv", "nondiff")`.
`round_digits`	A numeric value specifying digits for rounding the printed summary table. Defaults to 2.
`verbose`	Logical; if `TRUE`, outputs progress messages per battery. Defaults to `FALSE`.

Value

A data frame of class `c("straightliner", "data.frame")` containing the battery-level means for each requested measure (and the maximum SPV `"spv_max"` if `"spv"` is requested), returned invisibly.

References

Kim, Y., Dykema, J., Stevenson, J., Black, P., & Moberg, D. P. (2019). Straightlining: Overview of Measurement, Comparison of Indicators, and Effects in Mail–Web Mixed-Mode Surveys. *Social Science Computer Review*, 37(2), 214–233. doi:10.1177/0894439317752406

Examples

```
survey <- data.frame(
  item1 = c(5, 5, 3),
  item2 = c(5, 5, 2),
  item3 = c(5, 5, 1),
  item4 = c(5, 1, 4)
)
batteries <- list(first_two = c("item1", "item2"), second_two = c(3, 4))
straightlining_qset(survey, variable_list = batteries,
                    measures = c("mir", "qsd", "spv", "nondiff"))
```

Index

∗ **datasets**
    careless_dataset, 2
    careless_dataset2, 2

careless_dataset, 2
careless_dataset2, 2

evenodd, 3

irv, 4

longstring, *4*, 5

mahad, 6
matching, 7
maxIdenticalRating, 7, *10*, *20*
meanRootPairs, 8, *11*, *20*
mir, 10
mrp, 11

nondiff, 12, *20*

print.straightliner, 13
psychant, 13, *15, 16*
psychsyn, 14, *14*, *16*
psychsyn_critval, *14*, 15, *15*

question_sd, 16, *20*

rawnn, 17

scale_point_variation, 17, *19, 20*
spv, 18
stline, *12*, 19
straightlining, 20
straightlining_qset, 21